

OncoRelay3D: Foundation-Model-Assisted Reconstruction of 3D Tumor Cell-Expression Fields for Mechanism-Aware Directed Cell-Cell Communication

Zhitong Li^{1,2}, Yixuan Jiang^{1,2}, Ziming Liu^{1,2}, Chenjie Wu³, Ningfeng Que⁴, Chao Li^{2,5-7*}

1.College of Medicine and Biological Information Engineering, Northeastern University, Shen yang, China
2.School of Science and Engineering, University of Dundee, Dundee, UK
3.International College of Pharmaceutical Innovation, Soochow University, Suzhou, China
4.Department of Big Data and Biomedical AI, College of Future Technology, Peking University, Beijing, China
5.Department of Clinical Neurosciences, University of Cambridge, Cambridge, UK
6.Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, UK
7.School of Medicine, University of Dundee, Dundee, UK *Corresponding author: cl647@cam.ac.uk

CMMCAWorkshop



Abstract

Spatial transcriptomics (ST) is typically limited to discrete 2D sections, making directed 3D cell-cell communication inference challenging. We propose OncoRelay3D, a multimodal framework that reconstructs continuous 3D tumor cell-expression fields and infers mechanism-aware communication using H&E morphology and ST expression. Experiments on metastatic head and neck squamous cell carcinoma data demonstrate improved spatial interpolation and reliable biological interpretation of tumor-microenvironment interactions.

Introduction

Background

- 1.Spatial heterogeneity:** Within the same tissue, cells at different spatial locations exhibit significant differences in type, state, gene expression, and interactions.
- 2.Cell-cell communication:** Ligand-receptor (LR) analysis helps characterize molecular signaling between sender and receiver cells.
- 3.Multimodal tissue information:** H&E captures tissue morphology, while ST provides spatially resolved molecular states.

Challenges

- 1.Limited 3D sampling:** ST is acquired as discrete 2D sections, resulting in sparse cellular neighborhoods along the z-axis.
- 2.Missing directed 3D formulation:** Existing CCC methods do not explicitly model which cell communicates with which cell, through which LR pair, and under which 3D geometry.
- 3.Underused morphology:** Most CCC approaches do not fully exploit the complementary structural information provided by H&E histopathology.

Contributions

- 1.3DCCC formulation:** Models directed cell-cell communication on a reconstructed 3D cell-expression field.
- 2.Foundation-model-assisted interpolation:** Uses H&E morphology as structural guidance and ST as the primary molecular signal to reconstruct intermediate sections.
- 3.Cancer interpretation:** Aggregates 3DCCC outputs into pathway-level results and visualizes directed communication in 3D tumor tissue.

Method

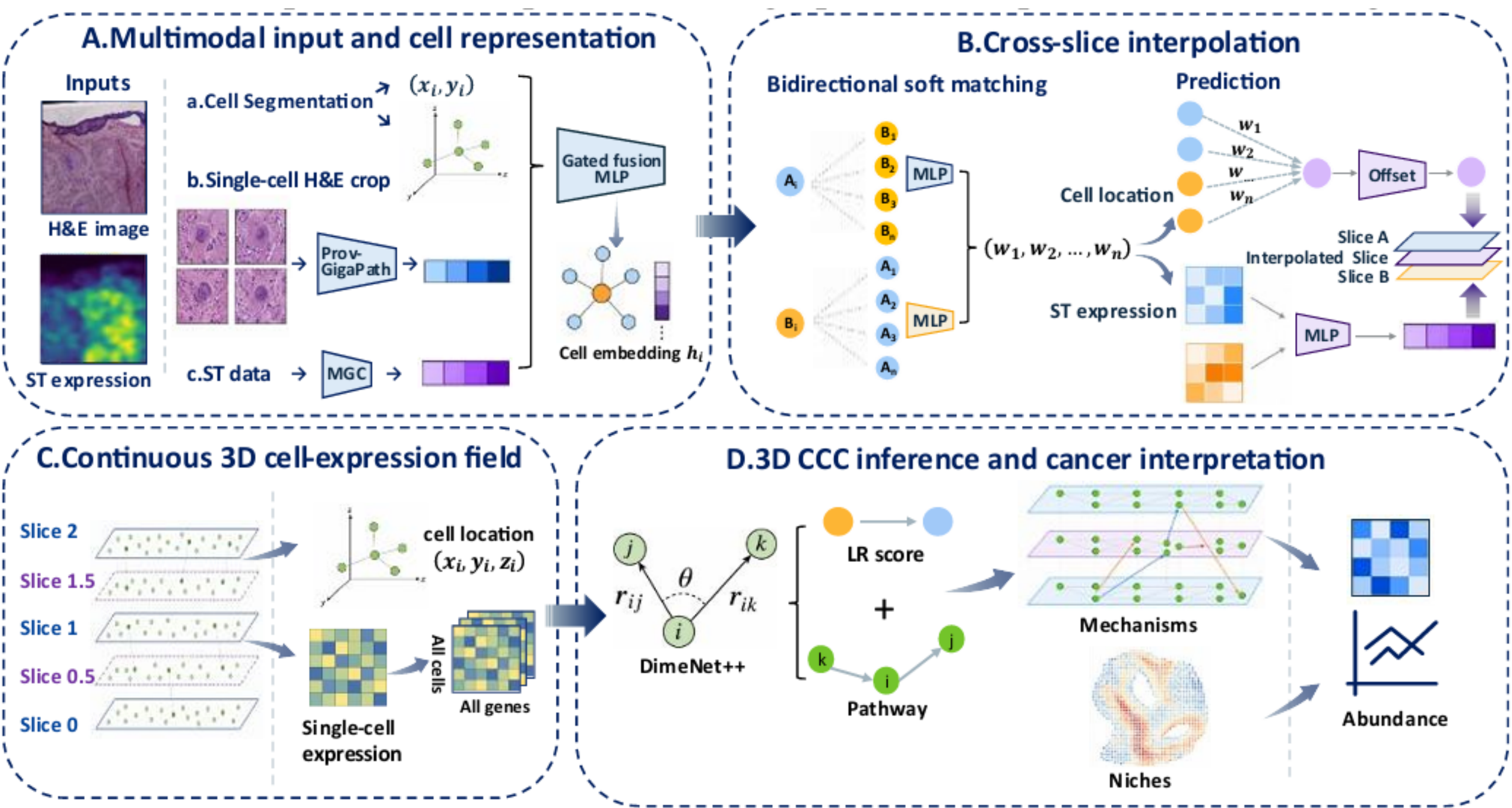


Figure 1. Overview of OncoRelay3D.

A.Cell-Level Multimodal Representation

- 1.Each cell is represented using its spatial coordinates, ST expression profile, H&E morphology features, and local cell-density information.
- 2.H&E provides morphology-aware structural information to guide cross-slice matching, while ST remains the primary source for molecular expression reconstruction.

B.Cross-Slice Interpolation

- 1.Reconstruct intermediate cell layers between two adjacent observed ST sections to reduce sparse sampling along the z-axis.
- 2.Candidate cells are identified using spatial proximity and feature similarity, followed by matching in both directions between adjacent sections.

C.Continuous 3D Cell-Expression Field

- 1.Original ST sections and reconstructed intermediate sections are integrated into a unified 3D representation.
- 2.Each cell retains its spatial location, expression profile, and source information.
- 3.The reconstructed layers increase cellular sampling along the z-axis and reduce discontinuities between observed sections.

D.Directed 3DCCC

- 1.Communication is modeled as directed edges from sender cells to receiver cells rather than as undirected cell associations.
- 2.Each communication edge explicitly considers ligand-receptor identity and the corresponding molecular signal.

Result

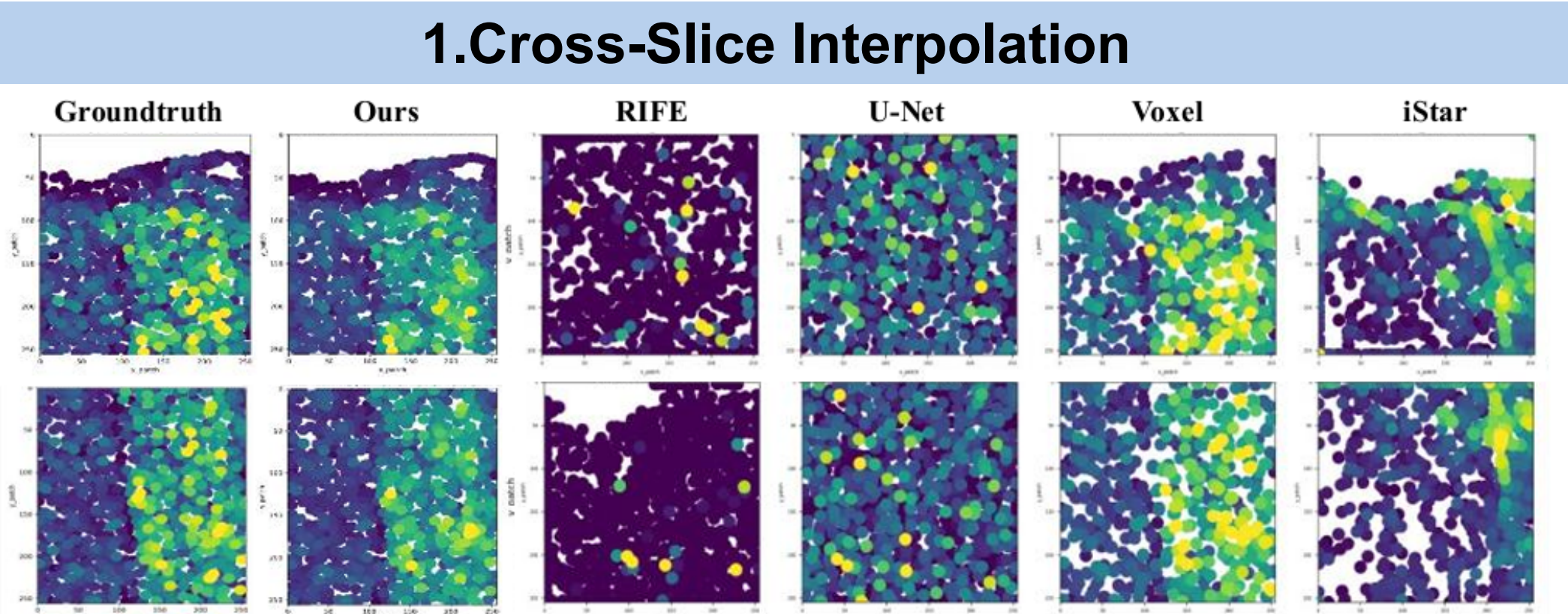


Figure 2. Cross-slice interpolation comparison.

Table 1. Cross-slice interpolation

Method	Expression	Spatial map	Location	Density	Topology	Overall
U-Net	0.025	0.322	0.000	0.000	0.448	0.159
RIFE	0.011	0.467	0.467	-0.001	0.467	0.282
Voxel	0.830	0.137	0.031	0.376	0.487	0.372
iStar	0.308	0.151	0.042	0.282	0.489	0.254
OncoRelay3D	0.825	0.934	0.670	0.639	0.883	0.789

OncoRelay3D achieves the highest overall interpolation score (0.789), with strong improvements in spatial structure, location, density and topology preservation.

2.Downstream 3DCCC Comparison

Table 2. Downstream 3DCCC comparison.

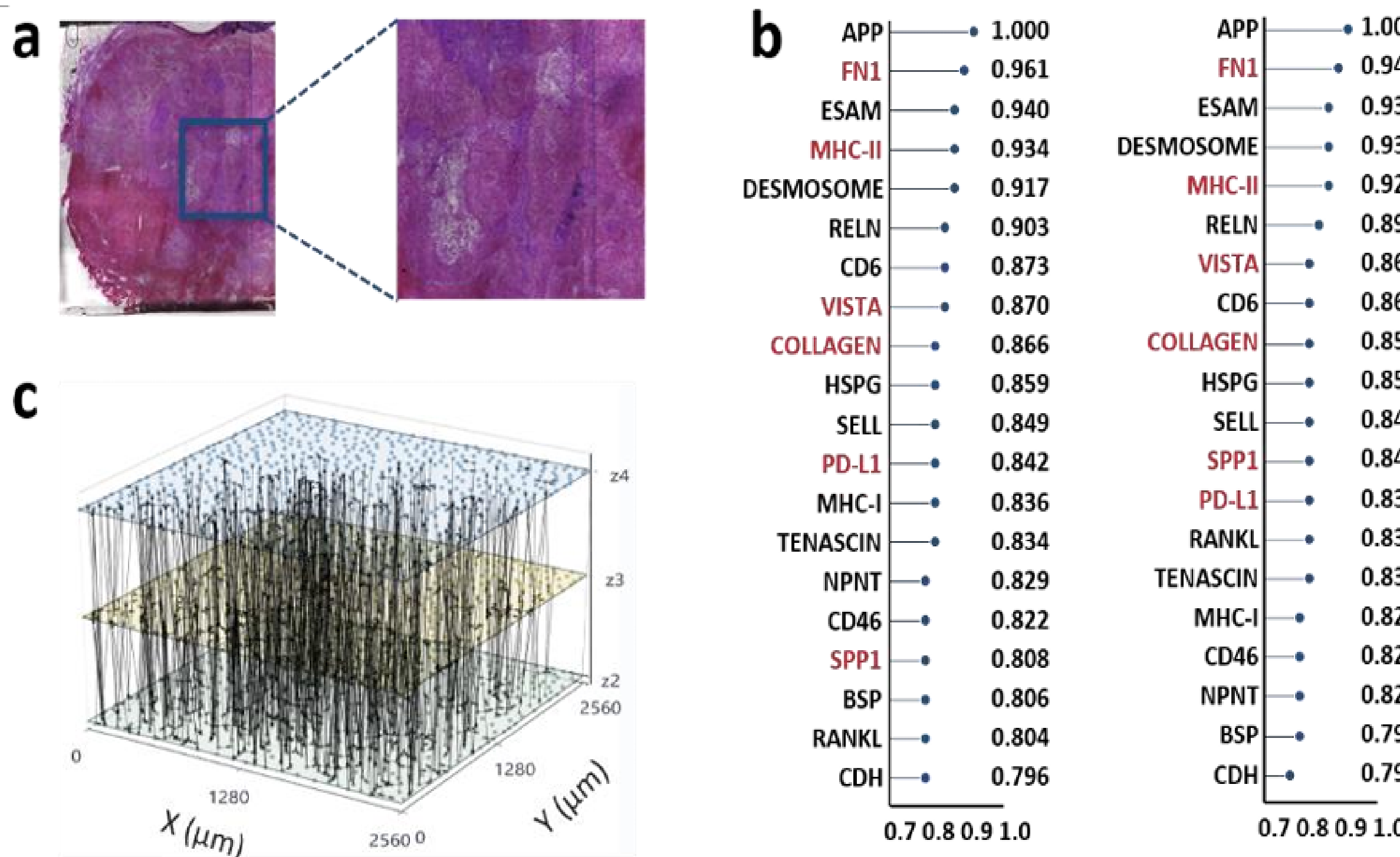
Method	LR	Direction	Mechanism	Overall
COMMOT	0.246	0.310	0.211	0.256
CellNEST	0.231	0.621	0.599	0.484
OncoRelay3D	0.605	0.515	0.723	0.614

1.In downstream 3DCCC inference experiments, we divided the evaluation metrics into ligand-receptor (LR) rationality, direction rationality, and mechanism rationality.

2. All compared methods were applied to the same reconstructed 3D cell-expression field to assess the overall biological consistency of the inferred communication network.

3. Through these experiments, our model achieved first place in LR rationality, mechanism rationality, and the overall 3DCCC score.

3.Cancer-Associated Pathway and 3D Communication Analysis



Discussion

Key Findings

- 1.Denser 3D representation:** Reconstructed intermediate sections increase cellular sampling along the z-axis and reduce discontinuity between observed ST slices.
- 2.Mechanism-aware communication:** Explicitly modeling LR identity, sender-receiver direction, 3D geometry and mechanism type improves biological consistency of inferred communication.

Biological Interpretation

- 1.Tumor Microenvironment:** Pathway-level analysis identifies tumor-microenvironment programs related to ECM remodeling, antigen presentation, and immune-checkpoint signaling.
- 2.Spatial Communication:** Directed 3D edges connect sender and receiver cells across reconstructed tissue layers. This links pathway-level signals to their spatially resolved cell-cell interactions.

Limitations

- 1.Generalizability:** Performance may depend on serial-section quality and cross-section alignment accuracy.
- 2.Future validation:** Additional serial-section cancer datasets are needed to evaluate robustness across tissue types and acquisition settings.

Conclusion

- 1.Reconstructs continuous 3D cell-expression fields from discrete ST sections.
- 2.Enables directed and mechanism-aware 3DCCC by integrating molecular, spatial and pathway information.
- 3.Supports cancer-mechanism interpretation through pathway-level analysis and 3D visualization of cell-cell communication.